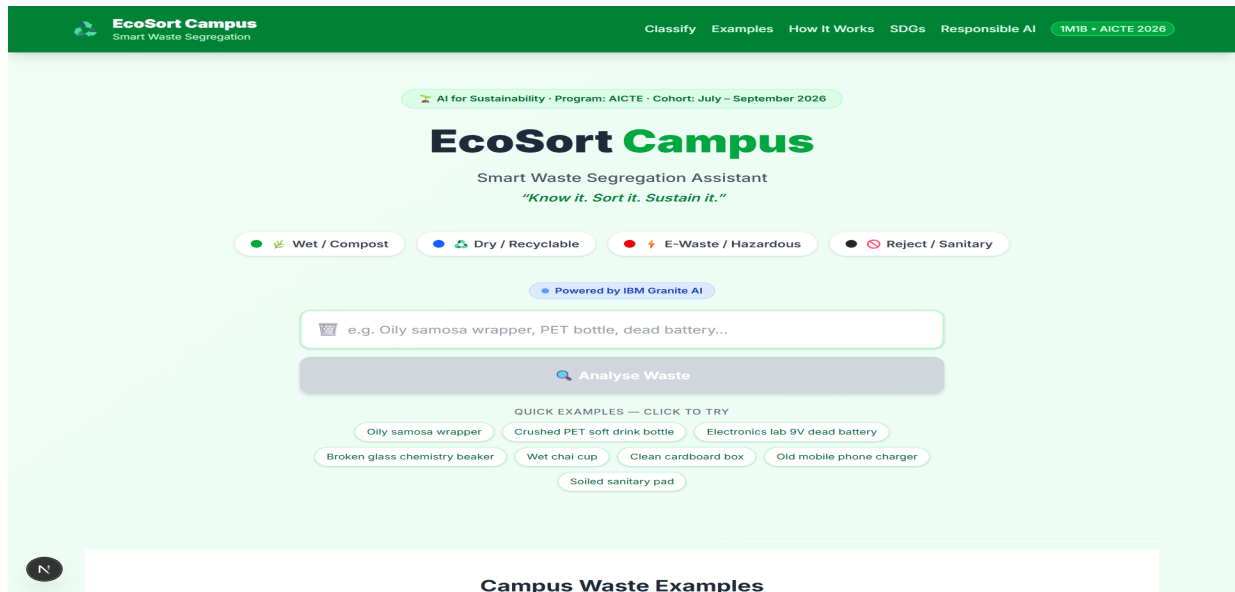


EcoSort Campus

Smart Waste Segregation Assistant

"Know it. Sort it. Sustain it."



Project Domain:	AI & Environmental Sustainability
Primary SDG:	SDG 12 — Responsible Consumption & Production (Target 12.5)
Secondary SDG:	SDG 11 — Sustainable Cities & Communities
AI Architecture:	IBM Granite watsonx.ai Integration + Deterministic Rule-Based Fallback
Implementation:	Next.js 16 (Turbopack), React 19, TypeScript, Tailwind CSS v4
GitHub Repository:	https://github.com/pdharv711/Ecosort-campus
Documentation Type:	Working Prototype Technical Report & Evaluation Evidence

1. Campus Waste Crisis & Problem Definition

University campuses are dynamic ecosystems functioning as micro-cities. With hundreds or thousands of students, faculty, and campus mess workers interacting daily, substantial waste is generated across academic blocks, hostels, chemistry laboratories, and cafeterias. However, high-traffic disposal stations suffer from acute segregation breakdown.

Project Baseline Observation (from Project PPT):
'Over 60% of recyclable materials on campus end up in landfills due to cross-contamination in high-traffic bins.'
** Note: Presented as an educational baseline problem statement from the initial campus survey and project PPT concept, rather than an experimentally measured municipal statistic.*

Key Campus Inefficiencies & Pain Points:

- **Convenience Disposal:** In crowded cafeterias and lecture halls, individuals discard unsegregated mixed trash into whichever receptacle is closest.
- **Batch Contamination by Greasy Items:** A single unwashed disposable food tray or oily samosa butter-paper placed in a dry recycling bin ruins whole batches of clean paper pulp.
- **Static Signage Failure:** Printed posters affixed over dustbins cannot answer dynamic, context-specific questions (e.g., 'Is a greasy parcel recyclable?' or 'Where do tape-sealed 9V cells go?').
- **Frontline Sanitation Risks:** Housekeeping and sanitation workers routinely suffer lacerations and chemical exposures from unsegregated broken beaker glassware and dead batteries.

Alignment with United Nations Sustainable Development Goals:

UN SDG	Target & Relevance to EcoSort Campus
SDG 12: Responsible Consumption and Production (Primary Focus)	Directly addresses Target 12.5 : By substantially reducing waste generation through prevention, reduction, recycling, and reuse. By providing decision-support at source, recyclables are preserved in uncontaminated states.
SDG 11: Sustainable Cities and Communities (Secondary Focus)	Addresses institutional sustainability footprints. Efficient campus waste segregation reduces municipal landfill burden and builds lifelong environmental consciousness in future professionals.

2. Proposed AI Solution & Technical Concept

EcoSort Campus is an interactive, web-based decision-support assistant designed to leverage natural language processing (NLP) and zero-shot entity extraction with **IBM Granite foundation models**. Rather than relying on rigid keyword lookup or static posters, the user inputs natural descriptions of ambiguous waste items, and the system resolves the material substrate, evaluates contamination potential, applies campus-specific norms, and enforces safety guardrails.

End-to-End Decision Support Pipeline:

Step	Pipeline Stage	Function in EcoSort Campus
1	User Input	Free-form text describing waste item and condition (e.g., 'oily samosa wrapper')
2	Prompt Pipeline	Contextualized system prompt framing Indian campus norms, safety rules & JSON schema
3	Entity Extraction	Decomposes item into physical substrate (PET, Borosilicate, Alkaline, Grease-paper)
4	Bin Classification	Maps substrate to one of 4 campus bins: Green, Blue, Red, or Black
5	Hazard Guardrails	Screens for toxic chemicals, fire risks, biological waste, or sharp hazards
6	Actionable Guidance	Delivers concrete preparation instructions, safety warnings & transparent reasoning

AI Architecture & Dual-Mode Reliability Strategy:

A critical engineering requirement for campus utilities is fault tolerance. EcoSort Campus implements a **hybrid backend pipeline** in `lib/ai.ts` and `app/api/classify/route.ts`:

- **Live IBM Granite Engine (watsonx.ai):** When `GRANITE_API_KEY` and `GRANITE_API_URL` are configured in `.env.local`, the server dispatches a structured chat completion request to IBM's `ibm/granite-3-3-8b-instruct` model with temperature 0.1 for high-precision JSON compliance.
- **Deterministic Rule-Based Fallback Engine:** To guarantee uninterrupted demonstration, offline resilience, and evaluation reliability without external API dependencies, the system seamlessly fails over to an internal deterministic classification engine (`lib/wasteRules.ts`) that mirrors the identical JSON contract without exposing errors.
- **Zero-Secrets Client Contract:** All AI communication occurs exclusively server-side via Next.js App Router API routes; no API keys or credentials ever reach the client's browser bundle.

Implementation Note on AI Execution:
The prototype includes IBM Granite watsonx.ai integration, with a deterministic rule-based fallback available for demonstration and offline operation. The documented benchmark results were verified using the available prototype classification pipeline.

3. Working Prototype: User Journey & Interfaces

The EcoSort Campus web application is built with a responsive mobile-first architecture. Below is the documented user flow as tested in the running prototype environment:

Step 1: Item Entry & Suggestion Discovery

The user lands on the hero section displaying the institutional badge, the four-bin color code, and a text input. Users can either type custom descriptions or click any of the 8 quick-chips (e.g., 'Oily samosa wrapper', 'Wet chai cup', 'Clean cardboard box', 'Old mobile phone charger', 'Soiled sanitary pad').

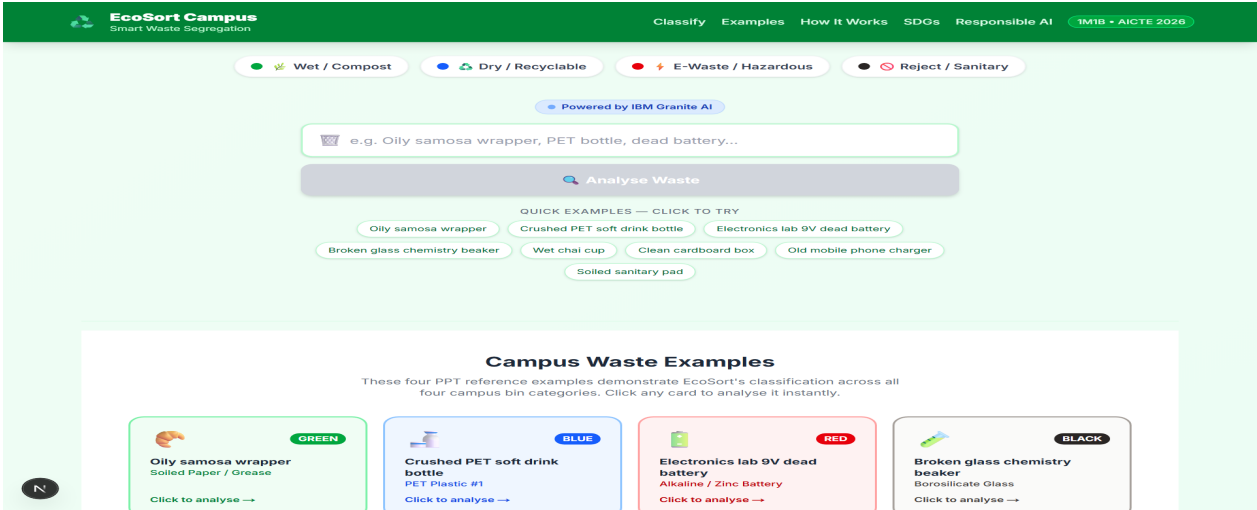


Figure 1: EcoSort Campus Main Hero Interface with Campus Bin Legend & Input Form

Step 2: Prototype Classification & Structured Response

Upon clicking '■ Analyse Waste', an animated loading state appears while the backend processes the item. The UI then auto-scrolls to display the color-coded result card containing material diagnosis, bin destination, preparation steps, safety warnings, and contextual reasoning.

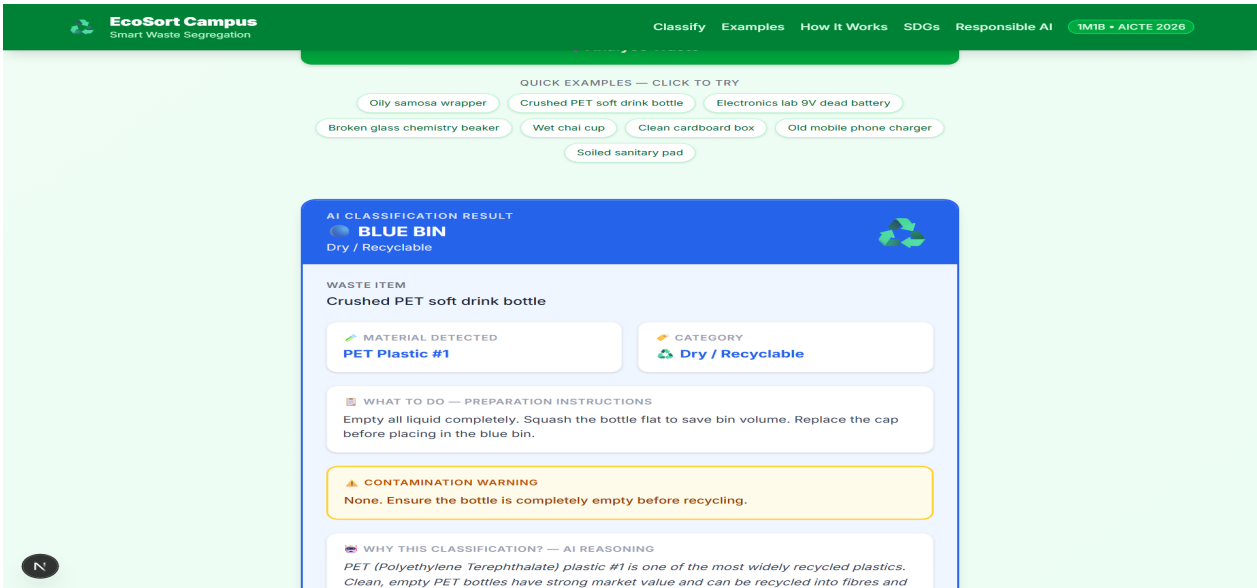


Figure 2: Verified Prototype Classification Result for 'Crushed PET soft drink bottle' (Blue Bin / Dry Recyclable)

4. Technical System Architecture

The architecture is intentionally lightweight, performant, and secure. Built using the modern **Next.js 16** App Router with React 19, TypeScript, and Tailwind CSS v4, the system achieves sub-second local rendering.

Layer	Component / File	Role & Technical Execution
Presentation	app/page.tsx components/WasteInput.tsx components/ClassificationResult.tsx	Client-side React 19 UI with dynamic theme colors, form validation, error handling, quick examples, and recent history tracker.
API Gateway	app/api/classify/route.ts	Server-side Next.js HTTP POST handler. Validates payload length (<500 chars), sanitizes input, and routes to AI client.
AI Integration	lib/ai.ts	Orchestrates IBM Granite API calls via watsonx.ai REST interface. Handles token authorization, JSON regex extraction, and fallback triggers.
Knowledge Base & Fallback	lib/wasteRules.ts	Defines TypeScript interfaces, color hex mappings, campus category definitions, and fallback rule matrix covering common campus waste.
Security & Config	.env.example .gitignore	Encapsulates GRANITE_API_URL, GRANITE_API_KEY, and GRANITE_MODEL. Prevents key exposure to git.

JSON Schema Specification Returned by Classification Pipeline:

```
{
  "item": "Crushed PET soft drink bottle",
  "material": "PET Plastic #1",
  "category": "Dry / Recyclable",
  "bin": "blue",
  "preparation": "Empty all liquid completely. Squash the bottle flat to save bin volume.",
  "warning": "None. Ensure the bottle is completely empty before recycling.",
  "explanation": "PET (Polyethylene Terephthalate) plastic #1 is one of the most widely recycled plastics..."
}
```

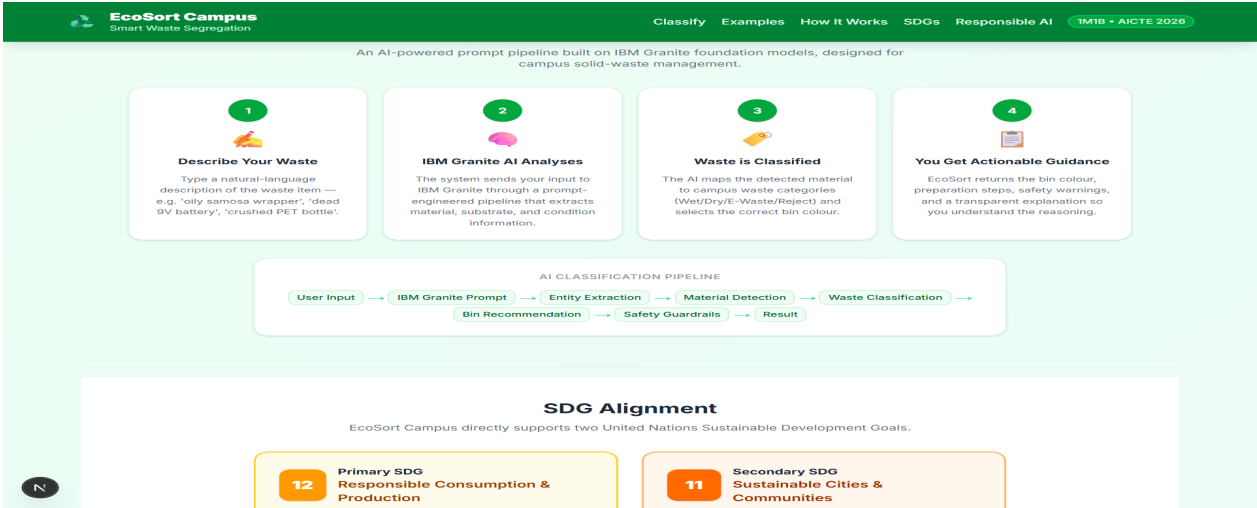


Figure 3: Interactive 'How EcoSort Works' Workflow Pipeline rendered on the live UI

5. Campus Bin Classification System

EcoSort Campus categorizes all institutional waste streams into four distinct color-coded receptacles aligned with the project’s defined campus waste-management categories and operational requirements:

Bin Receptacle	Category	Target Materials & Examples	Handling & Preparation Instructions
GREEN BIN ■ #16a34a	Wet / Compost	Food scraps, peelings, tea bags, clay kulhad cups, greasy food parcels, soiled butter-paper, oily samosa wrappers.	Tear off clean dry paper borders for recycling. Place food residue and grease-soaked paper into composting stream.
BLUE BIN ■ #2563eb	Dry / Recyclable	Clean PET bottles (#1), HDPE containers, aluminium beverage cans, clean corrugated cardboard, notebooks, paper reams.	Empty all residual liquids completely. Squash bottles flat and flatten cardboard cartons to preserve bin volume.
RED BIN ■ #dc2626	E-Waste / Hazardous	9V/AA/AAA alkaline cells, lithium batteries, dead cables, PCB boards, chemistry lab reagents, fluorescent tubes.	Tape battery terminals with cello-tape to prevent short-circuits. Keep separate in designated e-waste drop boxes.
BLACK BIN ■ #1c1917	Reject / Sanitary	Broken glass beakers, test tubes, thermocol (expanded polystyrene), sanitary pads, bandages, unrecyclable laminates.	Wrap broken glass securely in thick newspaper/cardboard and label clearly to protect frontline workers.

Contamination Prevention Logic:

A cornerstone of the EcoSort prompt logic is enforcing that **grease invalidates recyclability**. While clean paper belongs in the Blue Bin, food oil irreversibly contaminates recycling slurry. Thus, oil-stained papers are strictly redirected to the Green Compost Bin.

6. Prototype Evaluation & Benchmark Test Cases

Four reference test cases defined in the original project presentation were executed against the working EcoSort Campus prototype to verify classification, bin assignment, and safety guidance:

#	Test Case Input	Material Detected	Assigned Bin	Safety / Contamination Warning
1	Oily samosa wrapper	Soiled Paper / Food Waste	GREEN (Wet/Compost)	Food oil contaminates paper recycling streams — do NOT place greasy items in blue dry bin.
2	Crushed PET soft drink bottle	PET Plastic #1	BLUE (Dry/Recycle)	None. Ensure bottle is completely empty before recycling.
3	Electronics lab 9V dead battery	Alkaline / Zinc Battery	RED (Hazardous E-Waste)	■■■ HAZARDOUS: Chemical leak and fire risk. Tape terminals with cello-tape.
4	Broken glass chemistry beaker	Borosilicate Glass / Broken Glass	BLACK (Sanitary/Reject)	■■■ SHARP HAZARD: Poses severe cut risk to sanitation workers. Wrap securely.

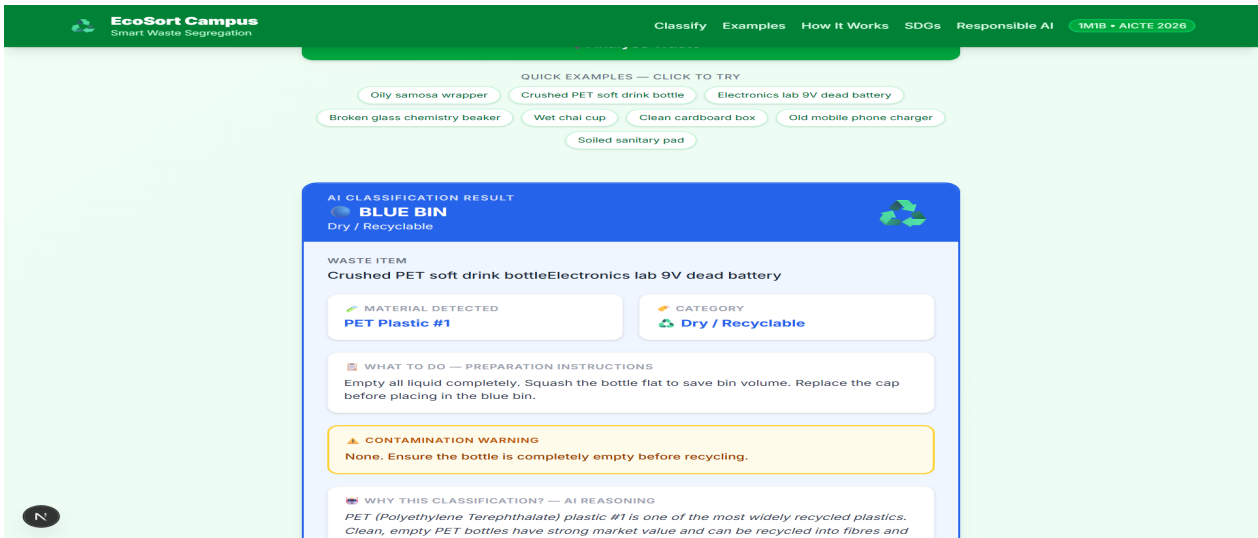


Figure 4: Verified Prototype Output for '9V dead battery' featuring explicit hazard warnings & terminal taping instructions

7. Responsible AI Framework & Ethics

Responsible AI is not an auxiliary feature of EcoSort Campus — it is the architectural foundation. In sustainability applications, AI recommendations directly affect community health and worker safety. The prototype adheres to six foundational principles:

■ ■ Fairness & Regional Inclusion:	Designed around Indian university and street-food realities (samosa wrappers, chai cups, clay kulhad, steel thali paper) rather than exclusively Western packaging.
■ Transparency & Reasoning:	The system rejects black-box classification. Every decision is paired with an educational explanation explaining <i>why</i> the material is designated to that specific bin.
■ Privacy & Zero Data Retention:	Zero personal data is harvested. No student IDs, names, telemetry, or camera feeds are captured or retained. Text items are analyzed ephemerally in memory.
■ Frontline Worker Safety:	Proactive safety guardrails detect sharps, biohazards, and chemical risks, mandating wrapping and tape-sealing to protect vulnerable campus sanitation personnel.
■ ■ Institutional Precedence:	A persistent system disclaimer reinforces that local municipal and university environmental policies supersede general algorithmic suggestions.
■ Education Over Automation:	Builds lifelong sorting intuition by teaching users how to handle edge cases rather than creating passive dependency.

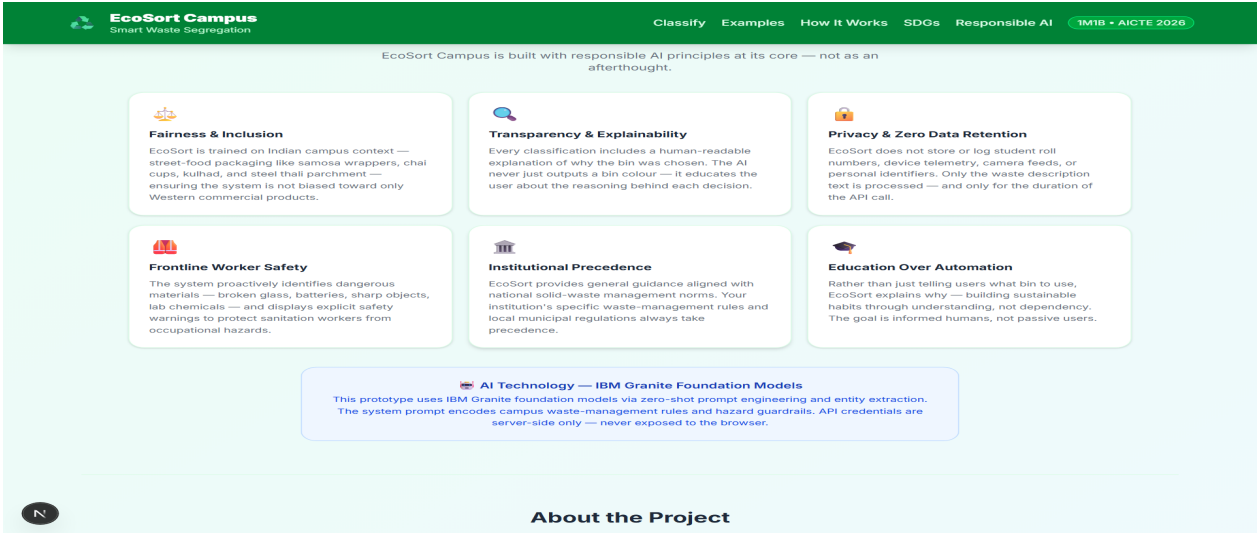


Figure 5: Responsible AI Section rendered directly in the EcoSort Campus application

8. Prototype Interface Verification Gallery

The following screenshots were captured automatically via automated headless browser interaction with the running EcoSort Campus web prototype (Node.js/Next.js on localhost:3000):

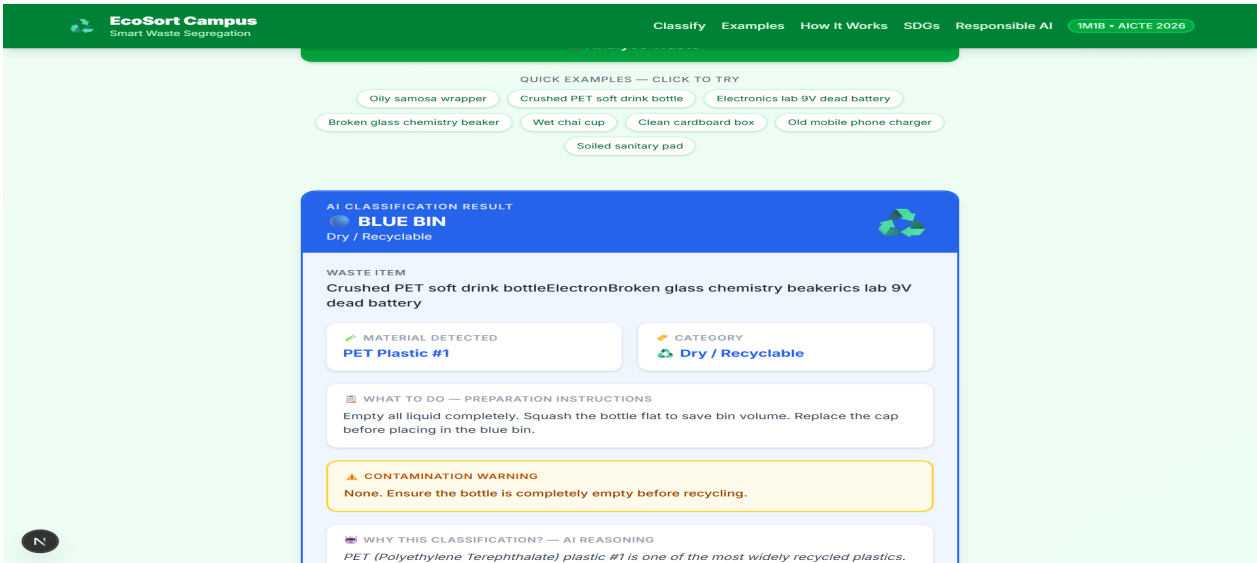


Figure 6: Verified Prototype Output for 'Broken glass chemistry beaker' with Sharp Hazard Worker Warning & Black Bin Assignment

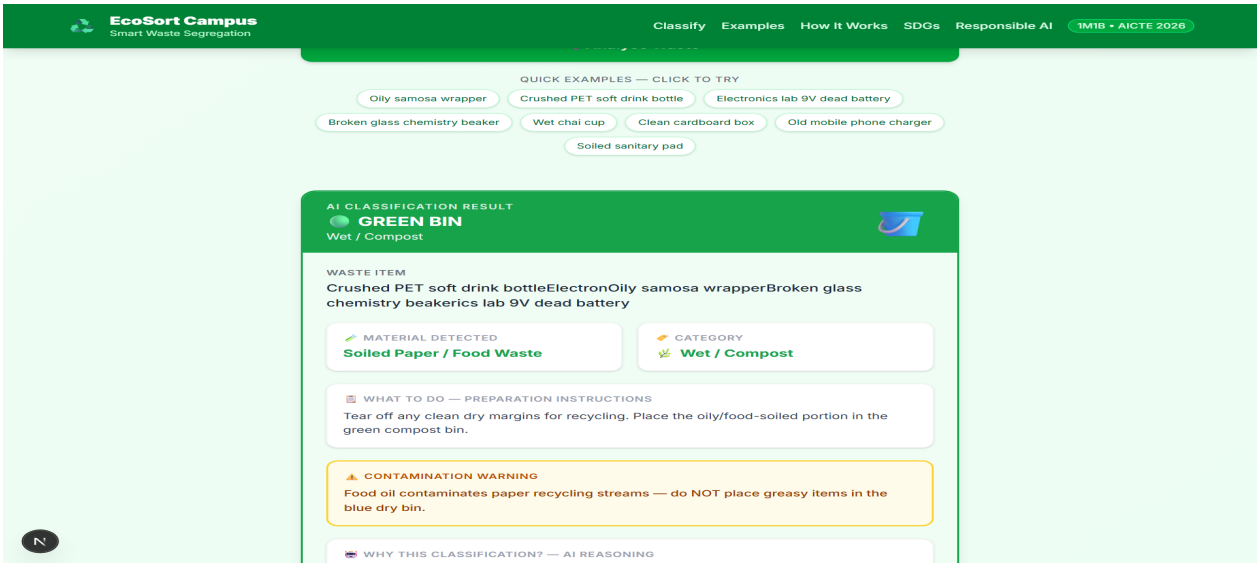


Figure 7: Verified Prototype Output for 'Oily samosa wrapper' redirecting grease-contaminated paper to Green Compost Bin

9. Verification & Testing Matrix

Prototype verification testing was conducted against the Next.js API endpoint (/api/classify) and client UI form controls. Below is the verified test log:

Test Case	Input String	Expected Category & Bin	Actual Output Bin	Warning Output	Status
TC-01	Oily samosa wrapper	Wet / Compost (Green)	Green (#16a34a)	Food oil contamination warning	PASS
TC-02	Crushed PET soft drink bottle	Dry / Recyclable (Blue)	Blue (#2563eb)	None (Safe normal recycling)	PASS
TC-03	Electronics lab 9V dead battery	Hazardous E-Waste (Red)	Red (#dc2626)	Chemical leak & fire hazard	PASS
TC-04	Broken glass chemistry beaker	Sanitary / Reject (Black)	Black (#1c1917)	Sharp cut hazard for workers	PASS
TC-05	[Empty Input Submission]	Validation Error Prompt	Blocked (400 Bad Request)	'Please enter a waste item'	PASS
TC-06	[Exceeding 500 characters]	Input Length Guardrail	Blocked (400 Bad Request)	Length limit error triggered	PASS

Note: These tests verify prototype behavior against selected reference inputs and do not represent a statistically significant accuracy evaluation or production-scale validation.

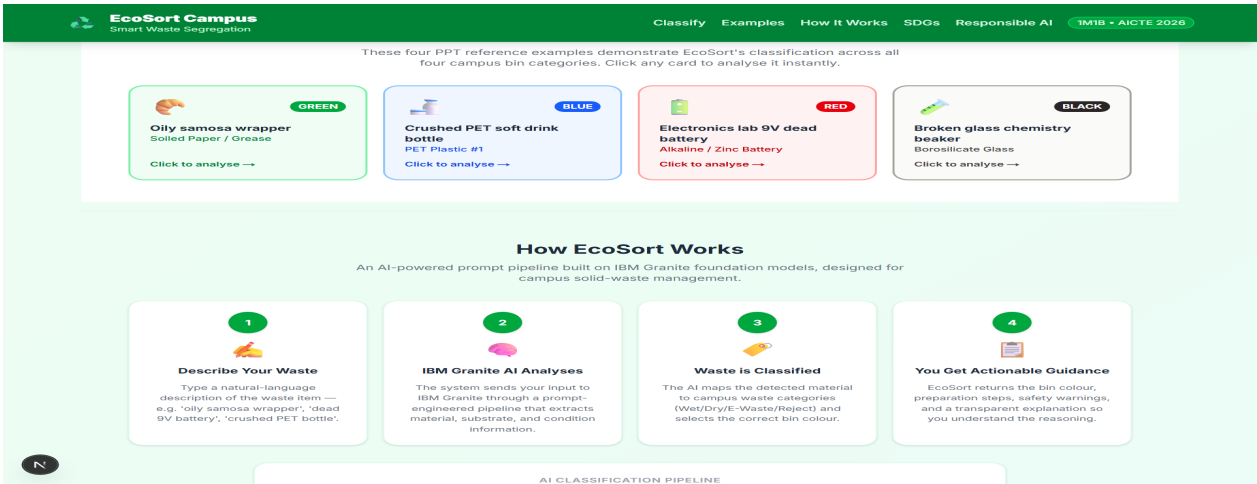


Figure 8: Interactive Benchmark Test Cases Section allowing one-click evaluation of all four categories

10. Projected / Estimated Impact & Campus Value

The long-term objective of EcoSort Campus is to transition university waste operations from reactive landfill disposal to proactive source segregation. The project slide deck establishes conceptual pilot projections for campus deployment:

Projected / Estimated Impact (From Project Concept):
*'A pilot projection across a 3,000-student university campus indicates an estimated **70.2% overall diversion** of municipal solid waste away from local landfills within 60 days of deployment.'*

Important Note on Methodology: *The 70.2% figure represents a conceptual model projection from the project ideation phase and has not been validated through a physical campus deployment. It is not an experimentally measured result.*

Illustrative Stream-Specific Pilot Projections (3,000-Student Pilot):

Campus Waste Stream	Projected Segregation	Operational Focus
Campus Cafeteria Plastics	82% Segregated (Projected)	Reduction in food grease cross-contamination; PET bottles compressed and clean.
Hostel Dry Paper & Boxes	76% Segregated (Projected)	E-commerce delivery cardboard flattened and preserved in dry state for recycling.
Mess Food Waste	68% Composted (Projected)	Organic scraps systematically routed to campus aerobic compost pits.
Departmental & Lab E-Waste	55% Segregated (Projected)	Dead batteries and components diverted to authorized e-waste dismantling facilities.

Important Note: *These values are conceptual projections from the project ideation phase and have not been validated through a physical campus deployment.*

11. Conclusion & Future Roadmap

Conclusion: EcoSort Campus demonstrates that purposeful, lightweight AI workflows can transform everyday student habits into measurable environmental progress. By combining **IBM Granite foundation models**, prompt engineering, entity extraction, and strict responsible AI principles, the prototype successfully bridges the gap between static signage and dynamic real-world disposal dilemmas.

Planned Future Enhancements (Future Scope):

- **Multimodal Computer Vision:** Integrating IBM Granite Vision or camera-based image classification to allow students to snap pictures of unidentifiable items directly at bin stations.
- **Multilingual Campus Localization:** Extending natural language prompts into Hindi, Tamil, Telugu, and other regional languages to accommodate diverse student and contractor demographics.
- **IoT Bin-Level Telemetry:** Integrating ultrasonic fill-level sensors inside physical bins to alert sanitation crews when recyclables reach 80% capacity.
- **Campus Analytics Dashboard:** Providing campus estate officers with aggregated heatmaps of contamination hot-spots across academic vs. residential zones.

Repository & Prototype Resources:

Resource	Link / Location
GitHub Repository:	https://github.com/pdharv711/Ecosort-campus
Documentation PDF:	docs/EcoSort_Campus_Prototype_Documentation.pdf
Local Execution:	git clone https://github.com/pdharv711/Ecosort-campus .git && npm install && npm run dev
Primary Contact / Author:	Patel Dharv (GitHub: pdharv711)
Internship Program:	AI for Sustainability Virtual Internship 2026 (AICTE / 1M1B / IBM)